
IK 510: Cognitive Neuroscience

Course Code	IK 510
Course Name	Cognitive Neuroscience
L-T-P-C	3-1-0-4
Intended for	3rd & 4th Year B. Tech, Masters and PhD
Approval	42nd Senate (Jan 2024)

Course Contents

- **Structure and function of nervous system** (5 hours)
 - Overview of nervous system
 - Brain & spinal cord anatomy
 - Cells of nervous system
 - Action potential
 - Neurotransmitters
 - Synapse
 - Networks
- **Sensation & Perception** (6 hours)
 - Olfaction
 - Gustation
 - Somatosensation
 - Audition
 - Vision
 - Sensation to perception
- **Attention** (4 hours)
 - Models of attention
 - Neural mechanism of attention
- **Attention control networks Action** (4 hours)
 - Anatomy & control of motor structures
 - Motor pathways & physiology
 - Motor planning, initiation and control
 - Motor learning and memory
- **Memory** (4 hours)
 - Learning and memory-anatomical & cellular basis
 - Mechanisms of memory
 - Memory consolidation
 - Memory deficits and amnesia
- **Emotions** (4 hours)
 - Defining & categorizing emotions
 - Theories of emotion generation
 - Neural systems of emotion processing
 - Emotion & cognition (learning & memory)
- **Language** (4 hours)

- Evolution & fundamentals of language
- Sensory, motor and cognitive aspects of language
- Theories of language comprehension & speech production
- Language deficits
- **Cognitive control** (4 hours)
 - Cognitive control anatomy, cross talks (sensorimotor) & deficits
 - Planning, Decision making & execution mechanisms
 - Cognitive control & neuropsychiatry
- **Social Cognition** (4 hours)
 - Theories of self
 - Understanding mental states of others (Theory of Mind-ToM)
 - Experience sharing & simulation theory
 - Mental state attribution
 - Neurobiology of social cognition
 - Social cognition deficits
- **Consciousness** (3 hours)
 - Mind-Brain problem
 - Consciousness theories- neuroscience perspective
 - Consciousness theories & practice-Yogic perspective
 - Miscellaneous-Plant & animal consciousness and others
 - Disorders of consciousness

Tutorials

- Methods in cognitive neuroscience
 - Behavioral experiments-tools & techniques
 - Introduction to neurophenomenology
 - Basics of bio signals
 - Electroencephalograph (EEG) & Heart rate variability (HRV)
 - Transcranial electrical current stimulation(tECS)
 - Transcranial magnetic stimulation (TMS)
 - Functional neuroimaging techniques (fNIRS & fMRI)
 - Introduction to computational neuroscience (ML & AI)

Text Books

1. Gazzaniga, Michael S., et al., **Cognitive Neuroscience: The Biology of the Mind**, W.W. Norton, 2018.

References

1. Kandel ER, Koester JD, Mack SH, Siegelbaum SA, **Principles of Neural Science**, 6th Edition, McGraw Hill, 2021